Brandon Jahoor

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I'm the guy who makes robots actually do things.

Education

University of Waterloo Ontario, Canada
Candidate for BASc in Mechatronics Engineering Honours, Minor in Computer Engineering

Technical Skills

Robotics & Perception: ROS2, Gazebo, Foxglove, OpenCV, kinematics/controls, RealSense, Nvidia Jetson, RPi
Software & Tools: Python, C++, LeRobot, VLA, Linux/Ubuntu, Isaac Sim & Lab, Docker, Agentic harnesses
Design & Prototyping: SolidWorks (CAD), 3D-printing, electrically/mechanically hands-on

Experience

Robotics Engineering Jan 2026 – April 2026
Thalassa Robotics *California*
- Created high fidelity internal simulation tool using **Isaac Sim** by integrating **controls, optics, hydrodynamics**, and maximising user friendliness and mimic real controls and physics beyond what Isaac Sim provides out of the box.

Robotics Research Assistant Sept 2025 – Dec 2025
University of Waterloo RoboHub *Ontario*
- Syncing **Xsens Awinda IMUs** with **7-axis robotic arm** (Emika Panda's) in to **mirror human arm motion**.
- Done in a ROS2 environment on Linux/Ubuntu, using **Docker** for replicability.

Robotics Engineering (Co-op) June 2025 – Aug 2025
University of Waterloo Robotics Team *Ontario*
- Developed **ROS2 Realsense** RGB & depth stream with compression and dynamic resolution selection; added **OpenCV** processing.
- Modelled/solved 6-wheel rocker-bogie drivetrain **forward & inverse kinematics** for mars rover running on **Nvidia Jetson**.

Robotics Engineering (Co-op) May 2025 – June 2025
BHF Robotics *Ontario*
- Implemented **ROS2 autodrive** routine; trained **deep learning AI model** for autonomous farming robot's weed/crop detection.

Projects

VLA Finetuning | *VLA, LeRobot, Nvidia Jetson, SO-ARM101, SmolVLA*
- Fine tuning SmolVLA to pick and place coloured blocks using LeRobot.

Chess Robot | *CAD, SO-ARM101, Controls, UX*
- Designed and prototyped custom robotic arm with UI and API capable of playing chess. Used LeRobot and Placo kin. solvers.

AI Roommate | *Nvidia Jetson, Ollama (Qwen2.5-0.5B) LLM, Ultralytics YOLOv8s Detection, REST API (FastAPI)*
- Developed scene-aware "AI Roommate" delivering real-time detection, assistance, & interaction from a functional dashboard.

Robot Arm Simulation | *Nvidia Isaac Sim & Isaac Lab, AI Policy Training, Reinforcement Learning*
- Setup Isaac Sim & trained RL policies for Franka & SO101 arms in (CUDA enabled) Isaac Lab; validated sim-to-real transfer readiness.

Hexapod Robot | *Nvidia Jetson (Jetpack), SO-ARM101 Robotic Arm, RealSense, AI Object Detection (OWL-ViT)*
- Led integration of Jetson-based OWL-ViT object detection on 18-DOF hexapod with SO101 arm for real-time autonomous following.

Autonomous Rover Simulation | *Gazebo, ROS2, Python, Navigation, AI Object Detection*
- Developed a Gazebo rover simulation with ROS2 navigation stack with obstacle avoidance for autonomous goal-seeking movement.

AI Object Detection | *AI, ROS2, Python, OpenCV, RealSense*
- Developed ROS2 RealSense camera package with compression for VLM object recognition with real-time OpenCV visualization.

WATonomous | *Docker, Foxglove, C++, LiDAR, A**
- Implementing ROS2 nav stack with path planning for autonomous navigation. LiDAR-based cost maps, and real-time odometry.

SWARM Robots | *Linux/Ubuntu, Docker, Gazebo, ROS2 Assembly, ESP32*
- Assembled research grade robots from scratch and deployed ROS2 and Gazebo containers in Docker to simulate swarm coordination.

3-DOF Robotic Arm | *ROS2, OpenCV, SolidWorks, PWM, 3D Printing, Raspberry Pi*
- Designed & 3D-printed using SolidWorks; colour-based token sorting using OpenCV HSV filtering for real-time classification.

Custom Agentic Harnesses | *Nvidia Jetson, llama.cpp, Hermes, Claude Code*
- Developed optimized llama.cpp local infr. powering Hermes superagents with Claude Code delegates for fully autonomous operation.